Lecture 13: Divisibility and Modular Arithmetic

Section 4.1

Section Summary

- Division
- Division Algorithm
- Modular Arithmetic

Division

Definition : If a and b are integers with $a \neq 0$, then *a divides b* if there exists an integer c such that $b = ac$.

- When a divides b we say that a is a *factor* or *divisor* of b and that b is a multiple of a.
- The notation $a \mid b$ denotes that a divides b.
- If $a \mid b$, then b/a is an integer.
- If a does not divide b, we write $a \nmid b$.

Example : Determine whether $3 \mid 7$ and whether $3 \mid 12$.

Properties of Divisibility

Theorem 1 : Let a, b, and c be integers, where $a \neq 0$.

i. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;

ii. If $a \mid b$, then $a \mid bc$ for all integers c;

iii. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof : (i) Suppose $a \mid b$ and $a \mid c$, then it follows that there are integers s and t with $b = as$ and $c = at$. Hence,

$$b + c = as + at = a(s + t). \quad \text{Hence, } a \mid (b + c) \quad \blacktriangleleft$$

Corollary : If a, b, and c be integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

Division Algorithm

When an integer is divided by a positive integer, there is a quotient and a remainder. This is traditionally called the "Division Algorithm," but is really a theorem.

Division Algorithm : If a is an integer and d a positive integer, then there are unique integers q and r, with $0 \leq r < d$, such that $\boldsymbol{a = dq + r}$.

- d is called the *divisor*.
- a is called the *dividend*.
- q is called the *quotient*.
- r is called the *remainder*.

Definitions of Functions **div** and **mod**

$q = a$ **div** d

$r = a$ **mod** d

Division Algorithm

Since, $a = dq + r$ and $q = a \text{ div } d \Rightarrow q = \lfloor a/d \rfloor$,

$r = a - dq \Rightarrow r = a - d * \lfloor a/d \rfloor$

Examples :

- What are the quotient and remainder when 101 is divided by 11?

  Solution : The quotient when 101 is divided by 11 is

  $9 = 101$ **div** 11, and the remainder is $2 = 101$ **mod** 11.

- What are the quotient and remainder when -11 is divided by 3?

  Solution : The quotient when -11 is divided by 3 is

  $-4 = -11$ **div** 3, and the remainder is $1 = -11$ **mod** 3.

Division Algorithm

- Prove that if a is an integer that is not divisible by 3, then $(a+1)(a+2)$ is divisible by 3.

  Solution : Assume that $3 \nmid a$.

  So, we get a non-zero remainder, $r \in \{1, 2\}$ when a is divided by 3. In other words, $a = 3q + 1$ or, $a = 3q + 2$.

  Where $q \in \mathbf{Z}$. Now,

  Case 1: $a = 3q + 1$. Then, $(a+1)(a+2) = (3q+2)(3q+3) = 3(3q+2)(q+1)$. Therefore, $3 \mid (a+1)(a+2)$.

  Case 2: $a = 3q + 2$. Then, $(a+1)(a+2) = (3q+3)(3q+4) = 3(q+1)(3q+4)$. Therefore, $3 \mid (a+1)(a+2)$.

Congruence Relation

Definition : If a and b are integers and m is a positive integer, then a is *congruent* to b *modulo* m if m divides $a - b$.

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m.
- We say that $a \equiv b \pmod{m}$ is a *congruence* and that m is its *modulus.*
- Two integers are congruent mod m if and only if they have the same remainder when divided by m.
- If a is not congruent to b modulo m, we write

$$a \not\equiv b \pmod{m}$$

Example : Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution :

- $17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.
- $24 \not\equiv 14 \pmod{6}$ since $24 - 14 = 10$ is not divisible by 6.

More on Congruences

Theorem 4 : Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

- If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$.
- Hence, there is an integer k such that $a - b = km$ and equivalently $a = b + km$.
- Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, $m \mid a - b$ and $a \equiv b \pmod{m}$. ◀

The Relationship between (mod m) and **mod** m Notations

- The use of "mod" in $a \equiv b \pmod{m}$ and $a \bmod m = b$ are different.
  - $a \equiv b \pmod{m}$ is a **relation** on the set of integers.
  - In $a \bmod m = b$, the notation **mod** denotes a **function**.
- The relationship between these notations is made clear in this theorem.

Theorem 3 : Let a and b be integers, and let m be a positive integer.

Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$. (*Proof in the exercises*)

Congruences of Sums and Products

Theorem 5 : Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
  - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
  - $b \cdot d = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$. ◀

Example : Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

Algebraic Manipulation of Congruences

- **Multiplying** both sides of a valid congruence by an integer preserves validity.

  If $a \equiv b \pmod{m}$ holds then $c \cdot a \equiv c \cdot b \pmod{m}$, where c is any integer, (Theorem 5 with $d = c$)

- **Adding** an integer to both sides of a valid congruence preserves validity.

  If $a \equiv b \pmod{m}$ holds then $c + a \equiv c + b \pmod{m}$, where c is any integer, (Theorem 5 with $d = c$)

- **Dividing** a congruence by an integer does **not** always produce a valid congruence.

Example : The congruence $14 \equiv 8 \pmod{6}$ holds. But dividing both sides by 2 does not produce a valid congruence since $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$.

Computing the **mod** *m* Function of Products and Sums

We use the following corollary to Theorem 5 to compute the **remainder of the product or sum of two integers** when divided by *m* **from the remainders** when each is divided by *m*.

Corollary : Let *m* be a positive integer and let *a* and *b* be integers. Then

$$(a + b) \pmod{m} = ((a \bmod m) + (b \bmod m)) \bmod m$$

and $ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$.

(*proof in textbook*)